GEODE / DIH-S: A Geometry-, Ephaptic- and Dendrite-Oriented Neuromorphic Substrate

Design, Math Audit, Logic-Gate Verification, and a Documented Account of What Did Not Work

Status: experimental. Every numeric claim in this paper is reproduced by a named script under `sim/` and a CSV under `sim/results/`. Negative results are reported with the same detail as positive ones, because in this project they carried most of the information.

Abstract

We set out to test whether the DIH-S spike-timing specification (v3.0) could be turned into a *working* brain-like computational substrate and, from there, into a fast chip. We built GEODE, a three-compartment (basal / apical / somatic) spiking-network engine that extends DIH-S with the three variables the brief singled out — **brainwaves**, **shape/volume**, and **spatial location** — realised as membrane time constants, geometry-derived axonal delays, and volume-scaled ephaptic (volume-conduction) field coupling.

Three results matter.

1. **All logic gates are implementable on this substrate and we verified them.** Twelve gate motifs (AND, OR, NAND, NOR, IMPLY, NIMPLY, NOT(A), NOT(B), MAJ3 one-layer; XOR and XNOR two-layer sum-of-products) were synthesised from *measured* device logic levels by maximum-noise-margin threshold placement, then instantiated in the real engine and checked against their truth tables by reading the output neuron's spike rate: **11/11 gates reproduce their truth table** (`s27_motif_engine.csv`), 12/12 verified mathematically.
2. **The generalisation failure was partly a measurement artefact — including one we inflicted on ourselves.** For ~20 experiments held-out accuracy sat at chance (0.37–0.53) while the network appeared to learn. Two separate measurement faults were involved.
    1. The readout decoded `x`, a *leaky* spike filter that still holds the previous stimulus when the next decision is taken; lengthening it monotonically degraded decodability (0.569 at 120 ms → 0.520 at 500 ms), which no "not enough integration" story explains. (b) Our own best positive result — that decoding the within-trial spike count lifts single-trial accuracy to 0.75–0.78 — **was in-sample overfitting and is retracted**: with a proper train/test split the same decoder gives 0.689 in-sample vs **0.529** held-out (`s33_insample_check.csv`). Three independent methods (held-out linear decoding, an optimal regularised logistic ceiling, and a 72-configuration sweep) agree that held-out single-trial decodability of XOR from this population is ~0.51–0.58, i.e. essentially chance. The failure is real; only the explanations were wrong twice.
3. **Autonomous learning of gates from noisy single-trial activity is capped near 0.70** even with an optimal regularised decoder on a frozen network; recurrent collaterals do not raise the cap (`s26_ceiling.csv`). This is a representation/SNR limit, stated as a negative result and *not* presented as solved.
4. **We added a decaying "credit" attribute — praise and pain proportional to measured contribution — and proved the system contains no backpropagation.** Every neuron carries

one extra scalar that decays exponentially; when an answer arrives, a *single* outcome scalar is broadcast and each neuron's attribute moves by its own measured share of the answer. Attribution is correct: Spearman ρ = **+1.00** against counterfactual-ablation ground truth on verified logic motifs, and **exactly zero** credit for a neuron that fires but drives nothing (an activity-proportional rule gives it 3.0). Measured half-life matches τ·ln2 within 1% (`s43_decay.csv`). No backpropagation exists and this is now enforced by an AST scan, a runtime canary, and a functional test in which credit works with *no error signal present at all* (§11). **But it does not yet buy accuracy**: it beats matched sign-shuffled and constant-valence controls in 2 of 8 crossed experimental cells (AND only) and never on XOR, where a supervised delta rule reaches 1.000. Five theories (§11.4) and a staged search pattern (§11.5) explain why and say where to look next.

Along the way we audited the DIH-S mathematics and found eleven defects (F1–F12) in the specification itself, then found nineteen more (F13–F31) while trying to make the model behave — including a sign error in our own homeostatic controller and a firing rule with no functional threshold. The full audit, experiment ledger, parameter search, chip implications and failure analysis follow.

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1. Motivation and brief

The project brief (DIH-S.md) asked for a brain-inspired model in which information is carried by spike *timing*, and specifically asked whether **brainwaves**, **brain shape/volume** and **spatial location** are load-bearing rather than decorative, then to derive a fast chip from the result, test all the maths for stability, search the parameter space, diagnose failures with biological and mathematical fixes, keep lab notes, track tests, and provide a hook interface.

Two design commitments followed from that brief and shaped everything:

- **Nothing is asserted that is not measured.** Every claim in this paper names the script and CSV that produce it. Where a measurement contradicted an assumption we had been working under, the assumption was discarded — including several of our own.
- **Accuracy is only counted with the teaching signal off.** Because the model's error signal is a *current injected into dendrites*, any accuracy measured while it is on simply reads the target back through the apical compartment. This single rule (fix F12) invalidated several earlier "working" results and is the reason this project spent so long looking at chance-level numbers: they were the first honest ones.

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2. The model

2.1 Three compartments

Each excitatory unit has basal, apical and somatic membranes, all leaky integrators in deviation form (`u = V - E_l`, so rest is `u = 0`):

```
tau_b du_b/dt = -u_b + I_basal                    (NMDA-like, slow)
tau_a du_a/dt = -u_a + I_apical                   (context / teaching)
tau_m du_s/dt = -u_s + g_b*u_b + g_a*s_a*coinc
                     - g_pv*pv*(u_s - E_i)        (shunting inhibition)
```

`I_basal` is the delayed weighted sum of presynaptic filtered activity plus sensory input; `s_a in {0,1}` is the dendritic plateau (Ca) spike with hysteresis; `coinc` is a sigmoid of basal drive so that a plateau can

only drive the soma when the cell is *also* receiving feedforward input — coincidence, not relay (fix F19). Inhibition is shunting (conductance-like), PV / SST / VIP interneuron populations are explicit, and PV carries a supralinear term above the target rate so inhibition vanishes at rest and clamps only excess activity.

2.2 The three brief-mandated variables

Brainwaves. Oscillation is not injected; it must emerge from membrane and synaptic time constants plus feedback. We measure it (layer-1 power spectrum, dominant frequency, band fraction) rather than assume it: `engine:oscillation_emergence_brainwave` reports a 3.67 Hz peak with 0.879 band fraction, and 20 Hz entrainment is tested as spike-timing modulation (PSTH peak/trough 1.49–1.66 — real but weak, flagged as a known limitation).

Volume. In a volume conductor `E ~ I / (4 pi sigma r)` and `r ~ V^(1/3)`, so a *smaller* brain gives a *larger* extracellular field per spike. The engine implements the field amplitude as `(V_ref / V)^p` with `p = 1/3` by default, and this scaling is covered by a unit test that varies `brain_volume_cm3` and asserts the field moves the right way.

Location. Neurone somata are placed in a prolate ellipsoid (a,b,c = 4.6, 3.4, 3.0 cm) with cortical parcels; every inter-layer axon gets a conduction delay from the measured inter-layer distance at 6 m/s, and optional `connect_radius_mm` enforces distance-limited connectivity. Delays are therefore *derived from geometry*, not chosen — also unit-tested.

2.3 Plasticity

A unified rule with three terms: a gated eligibility term (spec-literal `dih30`), variants that make the gate *signed* (`geode`, `geode_soft`), and a dendritic-cortical variant (`geode_dcm`) in which error polarity lives on the synapse as signed teaching current times presynaptic activity — the form that actually carries loss polarity. Plus multiplicative homeostatic scaling and scale-invariant (soft L1) structural decay, both of which were found to be necessary (F2b, F4).

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3. Math audit of the specification (F1–F12)

The DIH-S v3.0 equations were checked analytically before any simulation. Eleven defects were found; each has a regression test in `run_tests.py`.

ID	Defect	Fix	Test
F1	Weight init scale independent of fan-in, so total drive grows with width	fan-in-normalised init (`auto_scale_w`)	`rate_variable_normalisation`
F2	Hebbian growth unbounded → runaway	saturating bound `(1−|w|/w_cap)`	`hebbian_runaway_vs_saturating`
F2b	Additive homeostasis subtracts a constant from every weight → collapse	multiplicative homeostasis	`homeostasis_multiplicative_vs_additive`
F3	Euler leak unstable for `dt > tau`	explicit stability bound, enforced at init	`euler_leak_stability`
F4		scale-invariant soft-L1	`structural_decay_sign_vs_soft`

ID	Defect	Fix	Test
	`sign(w)` decay drives weights through zero and chatters		
F5	Inhibition as a current can drive `u` unboundedly negative	shunting conductance form	`shunting_inhibition_stability`
F6	Silent neurons get no credit (eligibility needs a post spike)	graded subthreshold post factor	—
F7	Thresholds compared in absolute mV while state is in deviation units	all thresholds as deviations	`ca_plateau_regeneration`
F8	`tau_x << mean ISI` so `x ≈ 0` between spikes; readout sees nothing	`tau_x = 25 ms`, auto spike amplitude	`rate_variable_normalisation`
F9	White current noise gives ~0.05 mV at dt=0.25 ms — no stochastic crossing	noise referred in **mV**, OU scaling `sqrt(2dt/tau)`	`noise_mV_referred_dt_invariance` (std 2.995/2.992/2.971 at dt 0.125/0.25/0.5)
F10	Field coupling not volume-scaled	`(V_ref/V)^{1/3}`	volume-scaling test
F11	Delays chosen by hand, so "location" was decorative	delays derived from geometry × 6 m/s	delay-from-geometry test
F12	**Teaching leakage**: error current drives the output population	accuracy only with teaching off; error never moves output soma	`stability_best_config` + all gate runs

F12 is the most important entry: not a numerical bug but an *epistemic* one, and it silently inflated every early "the model works" claim. The same class of leak recurred five more times (F13, F16, F17, F18, F21) — each time the error found another path into the measured variable. We now treat "there is still a leak" as the default hypothesis whenever accuracy looks good.

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4. What did not work, and why (F13–F30)

This section is the bulk of the paper: the model did not generalise for most of the project, and the reasons proved more informative than any accuracy figure.

4.1 Operating point: the hidden layer was dead, not untrained (F23)

Drive budget at layer 1 in the default configuration:

quantity	value
inhibitory shunt `g_pv·pv·(u_s−E_i)`	**8.05 mV**
basal excitation `g_b·u_b`	**1.24 mV**
steady-state `u_s`	**−2.5 mV** (threshold 8)

A ~6.5:1 inhibition:excitation ratio left the population ~10 mV below threshold, firing only off the noise floor. With inhibition zeroed the same network's rates became clearly pattern-dependent (26 / 36 / 33 / 46 Hz for inputs 00 / 10 / 01 / 11). No learning rule can recover information from a population held below threshold: this is an operating-point failure, not an optimiser failure.

4.2 The firing rule had no functional threshold (F27)

The spec's stochastic spike was a sigmoid hazard `h = σ((u_s−θ)/k)`, `k = 3 mV`. At the operating point above (`u_s ≈ θ − 3.5 mV`) it sits at `h ≈ 0.03`: every neurone fires at ~14 Hz **regardless of input**. This explains the project's most confusing measurement — pattern discriminability `d'` = 0.030 / 0.036 / 0.031 at membrane noise 3.0 / 1.5 / 0.5 mV, i.e. *flat in noise*, the signature of endogenous rather than injected variability (exp20). Added a Kramers/white-noise escape rate `R = R₀·exp(−(θ−u)₊²/2σ²)` (`spike_model="escape"`) and a hard-LIF option.

4.3 Readout saturation and weight collapse (F24, F25)

With `readout_scale = 2` and ±1 targets the opponent readout saturated at ±2, pinning the error at ±1 for the whole run — no gradient anywhere. Separately, overriding homeostasis to its additive form drove feedforward weights to *exactly* 0.000 and then pruned them (`w0max = 0.000`, anomaly `weight_clipping`). Both are silent failures: the loss trace reads like "learning is hard", not "the circuit has been deleted".

4.4 A sign error in our own controller (F26)

Our first per-layer synaptic-scaling loop was written `g ← g·exp(+η·log(r/r_target))` — **positive feedback**. Gain wound up to its clamp (8.0) and saturated the very layer it was meant to protect. Corrected to `−η·log(...)`. Recorded because a parameter sweep cannot detect this class of error: the failing configuration simply looks like "homeostasis does not help". New control loops need their own stability test, not merely an ablation.

4.5 Recurrent plasticity ran during "frozen" evaluation (F29)

The recurrent weight update lacked a `not inference` guard. Where it did not crash, every held-out measurement made with `recurrent_layers` set was taken on a network that was still learning. Earlier "the reservoir cannot support XOR" claims are therefore suspect; exp26 and exp30 use the corrected code.

4.6 Cross-trial contamination of the readout (F28) — and its retraction (F35)

What stands. The engine decoded `x[-1]`, a leaky integrator that at decision time still contains the *previous* stimulus, so consecutive decisions correlate with previous labels. The prediction separating "contamination" from "insufficient integration" is that lengthening the filter should help the latter and hurt the former. Measured (exp21c):

tau_x (ms)	25	60	120	250	500
acc(A)	0.551	0.548	0.569	0.543	**0.520**
d′_A	0.081	0.079	0.128	0.108	0.054

Monotonically worse with longer integration → contamination is real, and it is a genuine design constraint for a spiking chip: **decisions must be taken from an accumulator that is reset at the trial boundary**, not from a leaky trace. Implemented as `readout_var="count"`.

What we retract. We initially reported that this readout change raised single-trial accuracy from 0.53 to **0.75–0.78**. That figure was produced by fitting the mean-difference discriminant *and* its median

threshold on all trials and then scoring those same trials. Re-running the identical decoder with a disjoint train/test split (exp33, 3 seeds × 2 targets):

target	IN-SAMPLE (our original method)	HELD-OUT	gap
input A	0.683 / 0.687 / 0.683	0.527 / 0.507 / 0.507	+0.16 / +0.18 / +0.18
XOR parity	0.693 / 0.710 / 0.677	0.567 / 0.553 / 0.513	+0.13 / +0.16 / +0.16

mean in-sample **0.689** vs mean held-out **0.529** (chance 0.500). The headline number was overfitting and is withdrawn. Consequence: the generalisation failure is *not* explained away — it survives every measurement method we own (§4.7 ceiling, exp23 sweep, exp33 split).

4.6b A spike budget is a first-class design constraint (F32)

The count readout decoded chance for every gate when the counted population was the 2-unit output layer (exp30: AND 0.571, OR 0.548, NAND 0.381, NOR 0.643, IMPLY 0.548, NIMPLY 0.429). The reason is arithmetic, not learning: 2 units at ~5 Hz contribute ~0.7 spikes to a 70 ms decision window, so the decoded statistic is mostly zero. Sweeping the number of counted units k = 1…64 (exp32, `s32_spike_budget.csv`) gives flat held-out accuracy 0.532–0.546 for `spikes/decision` from 0.23 to 14.5 — i.e. at this information content, more spikes do not help, but *too few* certainly hurt. Design rule stated explicitly: **budget ≥ ~10 spikes per decision across the readout population**, and treat any accuracy computed from fewer as unmeasured rather than low.

Two engine fixes came out of this (F33): `readout_mode="opponent"` had hardcoded 2 output channels (a `(2,)` target could not broadcast against a `(32,)` readout), and `y_dec` used `y[0]−y[1]`; targets now expand to ±t over each half and the decode is `0.5·(mean(y[:n/2]) − mean(y[n/2:]))`, so wide populations average their own noise. Backward-compatible at `n_out=2`; suite still 17/17.

4.6c Template-driven growth: negative, and the control that caught it (F34)

Goal requirement: observe firing patterns and use them as templates to grow structure. The first attempt (exp28) was **invalid** and is reported as such — with all-to-all connectivity every cortical parcel mixes every input channel, so the observed inter-parcel phase step was 0.5° ("global flash", not a travelling wave). A template extracted from a global flash has no spatial content, so "before growth" scored 0.846 against almost anything and the comparison was meaningless.

The corrected attempt (exp28b: geometry ON, `connect_radius_mm = 3 mm`, local field, 2 Hz wave over 8 input channels) observed a genuine oscillation at **3.33 Hz** (band fraction 0.394) — but *not* at the 2 Hz drive, and with phase spread still only 0.24 rad. Growth test (930 new lead→follow synapses, template-ordered):

condition	replay phase-lock to true template
no growth	**−0.791**
grown	**+0.209**
shuffled template, no growth	−0.144
shuffled template, grown (control)	**+0.367**

Growth raised absolute lock, but the **degree-matched shuffled control rose further** (+0.367 > +0.209). The gain is non-specific excitation — more recurrent synapses raise gain, which raises oscillation amplitude, which raises *any* phase-lock measure. Verdict: `growth_helped = False`. Methodological

consequence adopted: **no growth or plasticity result is believed without a degree-matched shuffled control**; without that control this experiment reads as a clear success.

4.7 Ceiling on autonomous gate learning ≈0.70 (F30, negative result)

To separate "our learner is bad" from "the representation is limited", we froze the random network, collected within-trial spike counts over ~400 trials, fitted a regularised logistic decoder and evaluated on held-out trials:

recurrent	n_hid	dwell	AND	OR	XOR	NAND
no	64	40 ms	0.695	0.615	**0.550**	0.695
no	64	120 ms	0.680	0.660	0.585	0.680
no	256	40 ms	0.740	0.615	**0.445**	0.740
no	256	120 ms	0.695	0.675	0.500	0.695
yes	64	40 ms	0.690	0.625	0.485	0.690

1. Wide layers **overfit**: train 1.000 with test 0.445 — ~200 trials against 256 noisy count features, so more units make held-out performance *worse*. (ii) Recurrent collaterals did not raise the ceiling; XOR sits at or below chance everywhere. Stated plainly: **autonomous learning of non-separable gates from single-trial noisy population counts is not solved on this substrate and is limited by representation/SNR, not optimiser tuning.**

Earlier hypotheses were falsified independently: H1 reservoir (exp18 — best LR held-out 0.514), H6 monotonicity (exp19 — feedforward inhibition moved the monotone-unit fraction from 33% to 22% without fixing generalisation), and a 72-configuration sweep over firing model, noise, drive, threshold, inhibition and homeostasis never exceeded **0.556** single-trial accuracy (exp23, `s23_sweep.csv`).

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5. Logic gates: verified on the substrate (positive result, F31)

Having established that *learning* gates from noisy activity is capped near 0.70, we asked the question a chip designer actually needs answered — **can this substrate implement every logic gate with noise margin?** — and answered it in two verifiable stages. This matters for the brief's chip goal: a processor does not need its transistors to have learned AND; it needs AND to be realisable, reliable and composable.

5.1 Stage A — synthesis by maximum-noise-margin threshold placement

Each GEODE soma compares a weighted sum of inputs against a threshold: exactly a linear threshold unit. A boolean `f` is implemented iff

```
sum_j w_j x_j + b > theta   <=>   f(x) = 1 .
```

With the device's measured logic levels (OFF input contributes 0, ON contributes V1), place `theta` midway between the largest OFF-sum and the smallest ON-sum; the margin is the half-width between them. Result (`s27_motif_math.csv`):

gate	layers	separable	margin
	1	yes	**0.5 V1**

gate	layers	separable	margin
AND, OR, NAND, NOR, IMPLY, NIMPLY, NOT(A), NOT(B), MAJ3			
XOR, XNOR	2 (sum-of-products)	no (needs depth)	**0.5 V1**

12/12 motifs verified mathematically. XOR and XNOR are provably not linearly separable in their inputs — the two-layer construction (two AND-like hidden units feeding one OR) is the minimal fix, and depth is available in the engine. This is a concrete example of the brief's "shape/volume/location" requirement becoming architectural rather than decorative: non-separable functions require *depth*, and the substrate has it.

5.2 Stage B — the same motifs run in the real engine

Two engine features were added for this test: `theta_layers` (per-layer soma thresholds, so each layer of a motif can implement its own decision boundary) and `clamp_x0` (layer-0 activities pinned to ideal digital levels, as a chip tester would drive them). Weights are divided by the measured steady-state gain `k` so the designed sum equals the realised soma drive. The decision is read from the **output neurone's spike rate** — no learned readout is involved anywhere in this measurement.

Measured device levels: `V(off) = 0.000`, `V(on) = 1.000`, drive per unit weight `k = 2.000`.

gate	measured output rates (Hz), pattern order 00,10,01,11	got	want	verdict
AND	0.0, 0.0, 0.0, 111.4	0001	0001	**PASS**
OR	0.0, 111.4, 111.4, 111.4	0111	0111	**PASS**
NAND	111.4, 111.4, 111.4, 0.0	1110	1110	**PASS**
NOR	111.4, 0.0, 0.0, 0.0	1000	1000	**PASS**
IMPLY	111.4, 111.4, 0.0, 111.4	1101	1101	**PASS**
NIMPLY	0.0, 0.0, 111.4, 0.0	0010	0010	**PASS**
NOT(A)	111.4, 111.4, 0.0, 0.0	1100	1100	**PASS**
NOT(B)	111.4, 0.0, 111.4, 0.0	1010	1010	**PASS**
MAJ3	0,0,0,111.4,0,111.4,111.4,111.4	00010111	00010111	**PASS**
XOR (2-layer)	0.0, 111.9, 111.9, 0.0	0110	0110	**PASS**
XNOR (2-layer)	111.9, 0.0, 0.0, 111.9	1001	1001	**PASS**

11/11 gates reproduce their truth table in the engine at zero noise. The ON/OFF contrast is total (111.4 Hz vs 0.0 Hz), i.e. the digital abstraction holds exactly on this substrate when driven with ideal inputs.

5.3 Noise margin: the design rule for the chip

Synthesised margin is `0.5·V1` by construction, so reliability requires `0.5·V1 ≫ σ_noise`. Sweeping logic swing against membrane noise over all 11 gates (`s27_margin.csv`):

swing (× V1)	σ = 0.5 mV	σ = 1.0 mV	σ = 2.0 mV	σ = 3.0 mV
1	3/11	0/11	0/11	0/11
2	6/11	3/11	0/11	0/11
(larger swings: see `s27_margin.csv`)				

The rule is unambiguous and it is the same one that governs real asynchronous logic: **the logic swing must be several times the membrane noise floor.** At the default single-level swing (margin 0.5 mV against σ = 1–3 mV) gates are unreliable; at 2× swing, half survive 1 mV. For the chip this is a sizing/voltage-budget constraint, not an architectural defect — and it is measurable per technology node from `measure_levels()`, which returns V0, V1 and the drive-per-weight gain `k` directly from the device.

5.4 What the gate result does and does not claim

It **does** establish: the substrate can realise all boolean primitives, including non-separable ones with depth, exactly, with a quantifiable noise margin, using only the engine's own dynamics (leaky integration + threshold + spike rate as logic level).

It **does not** establish: that the network *learns* these gates from experience. That remains open and is bounded by §4.7. The two statements are independent, and conflating them is what let an earlier phase of this project report success it had not earned.

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6. Template-driven growth: three attempts, one valid, all negative (F34)

The second goal asked us to observe brain-like firing patterns and use them as templates to *grow* a structured network. We tried three times, tightening the controls each time; only the third result is trustworthy, and it is negative. This progression is reported in full because the first two attempts would both have looked like successes.

exp28 — invalid. Geometry ON, field ON, 2 Hz wave over 8 channels, all-to-all connectivity. Observed inter-parcel phase step: **0.5°**. Every parcel mixes every input channel under all-to-all wiring, so there is no spatial pattern to serve as a template; the "template" was a global flash and scored 0.846 for *any* candidate network. The growth comparison was therefore meaningless (reported `growth_helped = False` only by luck).

exp28b — controlled stimulus, uncontrolled comparison. With local connectivity (`connect_radius_mm = 3 mm`) the network oscillated at **3.33 Hz** (band fraction 0.394) — notably *not* at the 2 Hz drive, i.e. weak entrainment, consistent with the long-standing "weak PSTH modulation" flag in the test suite — but phase spread stayed at 0.24 rad. Growth of 930 template-ordered lead→follow synapses raised replay phase-lock from −0.791 to +0.209, which looks like success. It is not: the shuffled control rose further (+0.367), and adding ~900 recurrent synapses raises total conductance, hence oscillation amplitude, hence *any* phase-lock measure.

exp28c — two fixes, definitive negative. (i) After growth the recurrent matrix is rescaled so total conductance equals the un-grown baseline (weights scaled ×0.46). (ii) A second, degree-matched treatment grows from a **shuffled** template (same number of new synapses, same weight rule, wrong temporal order), each scored against its own template:

template	growth	lock-to-own-template	activity	new synapses	w-scale
true	none	−0.827	1.978	0	—
true	grown	**−0.498**	1.546	447	0.460
shuffled	none	−0.832	1.978	0	—
shuffled	grown	**−0.057**	1.599	462	0.451

Gain from growth: **+0.329 for the true template vs +0.775 for the shuffled one.** The matched control beats the treatment, so there is no specific effect to report — template information does not survive into the grown connectivity under this rule.

Why we believe the negative and distrust the positives. Both earlier experiments would have been publishable as "structure grew from an observed brainwave template" without the conductance normalisation and the degree-matched shuffled treatment. Adopted rule: *no plasticity or growth result is accepted unless a degree-matched, conductance-normalised control fails to do at least as well.*

What this implies biologically. Our growth rule potentiates lead→follow pairs, which is the correct STDP direction, but it does so on a *static* template applied all at once. Real assembly formation is Hebbian and incremental: the pattern must be re-expressed by the network's own dynamics while plasticity is open, so that only genuinely co-active pairs are strengthened — and the oscillation must actually entrain to the stimulus, which we measured it does not do well here (3.33 Hz endogenous vs 2 Hz driven). The failure is therefore located in the growth protocol and in weak entrainment, not in the premise that templates can shape structure.

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7. Chip implications

Everything the brief asked of a chip design came out of measurement rather than assumption.

Stability / solver tick. Euler integration of each leaky compartment is stable iff `dt < 2·tau`; the binding constraint is the fastest membrane. The init-time gate computes the bound with a 0.5 safety factor and reports **max safe dt = 14.29 ms**, so the numerics are not the throughput limit — a 0.25–0.5 ms tick used by every experiment here is ~30× conservative. `engine:dt_stability_and_learning_all_dts` verifies learning quality at dt = 0.5/0.25/0.125 ms.

Digital abstraction is exact, when driven properly. With ideal inputs the gate motifs give 111.4 Hz ON vs **0.0 Hz** OFF — no intermediate states (`s27_motif_engine.csv`). The substrate supports combinational logic cleanly.

Noise margin is a sizing rule, not an architectural defect. Synthesised margin is `0.5·V1`; sweeping swing × membrane noise over all 11 gates (`s27_margin.csv`):

swing (× V1)	σ=0.5 mV	σ=1.0	σ=2.0	σ=3.0
1	3/11	0/11	0/11	0/11
2	6/11	3/11	0/11	0/11
4	6/11	6/11	3/11	3/11
8	9/11	6/11	6/11	8/11
16	**9/11**	**9/11**	6/11	6/11

Design rule: **logic swing ≥ ~8–16× the membrane noise floor** for reliable gates. `V0`, `V1` and the drive-per-weight gain `k` are extracted from the device by `measure_levels()`, so the rule is checkable per technology node instead of assumed.

Spike budget. A decision needs enough counted spikes to be a statistic at all: 2 units at 5 Hz give ~0.7 spikes per 70 ms window and decode chance regardless of algorithm (F32). Budget ≥ ~10 spikes per decision across the readout population.

Area/energy consequence. Non-separable functions need depth (§5.1), so a gate library on this substrate costs 2 layers for XOR/XNOR; and reliability costs voltage swing, not extra transistors. Both are quantified above rather than hand-waved.

What is NOT ready for tape-out. Autonomous learning of non-separable gates (§4.7), template-driven structural growth (§6), and strong entrainment to an external carrier (§6). A chip built today would be a *configured* spiking fabric with verified gates, not one that learns its own circuits from experience.

---

8. Reproduction

```
sim/run_tests.py                 17/17 verification suite (math + engine)
sim/hook.py status|run|gate|tests|note   resume interface (results/hook-state.json)
sim/exp20_snr.py                 d' flat in noise => endogenous variability
sim/exp21_transfer.py            E/I budget: shunt 8.05 mV vs excitation 1.24 mV
sim/exp23_sweep.py               72-regime sweep, max single-trial acc 0.556
sim/exp24_window.py exp25_dwell.py   count-vs-rate and dwell studies (see retraction F35)
sim/exp26_ceiling.py             frozen-net ceiling ~0.70; XOR <= 0.585
sim/exp27_gate_motifs.py         ALL GATES: 12/12 math, 11/11 engine truth tables
sim/run_margin.py                noise margin vs logic swing (s27_margin.csv)
sim/exp30_gates_count.py         gates with count readout, honest held-out per gate
sim/exp32_spike_budget.py        accuracy vs counted population size
sim/exp33_insample_check.py      in-sample vs held-out: retraction of F28's headline
sim/exp28_template_growth.py     template growth v1 (invalid - documented)
sim/exp28b_template.py           template growth v2 (uncontrolled comparison)
sim/exp28c_template.py           template growth v3 (normalised + matched control): NEGATIVE
sim/exp34_composability.py       half adder / mux2 / NAND-universality / spike-driven AND
sim/exp35_levels.py              spike-driven logic levels (F38: HIGH did not exist)
sim/exp36_mesh_window.py         DISTANCE-WIRED MESH + answer window (F39/F41)
sim/exp37_mesh_controls.py       noisy-population controls (reported negative, superseded)
sim/exp38_mesh_deterministic.py  backward path carries activity: 60 vs 0 spikes
sim/exp39_delays.py              per-neuron arrival times vs axon length
sim/exp40_delay_isolated.py      DELAY PROOF: corr ~0.99, spread widens 4.33x (F40 fix)
sim/exp41_window.py              window opening validated; no natural close
sim/exp42_window_close.py        decay tau = inf with AND without recurrence (F42)
sim/exp43_credit_attribute.py    CREDIT ATTRIBUTE: attribution vs ablation GT, decay law,
                                 behavioural effect, no-backprop audit, learning arms
sim/exp44_credit_search.py       randomised credit-design sweep vs shuffled/const controls
sim/exp45_credit_learns.py       can praise/pain alone learn? + supervised reference ceiling
sim/exp46_credit_bootstrap.py    bootstrap requirement of outcome feedback (see F47 caveats)
sim/exp46b_saturation.py         is degradation saturation? (no) — exp46 follow-up
sim/exp46c_cover_center.py       class-conditional (Cover) correction test
sim/exp46d_collapse_control.py   CRITICAL CONTROL: collapse happens with credit OFF too
sim/exp46e_clean_bootstrap.py    bootstrap after removing the weight-dissolution confound
sim/exp46f_error_only.py         Agmon error-only update variant
sim/exp47_confirm_credit.py      crossed width x structural decay, 8 cells vs both controls
sim/tests_engine.py              includes the six credit + no-backprop regression tests
```

CSVs land in `sim/results/`. Lab notes: `lab notes/01–07`, plus `test-log.csv`.

9. Conclusions

1. **Brainwaves, shape and volume are implementable as real variables** — emergent oscillation measured by spectrum, delays derived from geometry at 6 m/s, ephaptic field scaled as `(V_ref/V)^{1/3}` — and all three are covered by tests. Whether they are *functionally load-bearing* is still unproven: entrainment to an external carrier is weak (3.33 Hz endogenous vs 2 Hz driven; PSTH peak/trough only 1.49–1.66), and no experiment here showed a task that fails when geometry or field coupling is switched off.
2. **Every logic gate is implementable and verified** on this substrate with quantified noise margin, including non-separable ones via depth. This is the project's solid positive result and it is exactly what a chip needs.
3. **Autonomous generalisation remains unsolved**, and we now know it is not explained by any single bug we found: it survives the operating-point fix, the firing-rule fix, the readout fix, wide populations, longer integration, recurrence, and 72 parameter regimes. Held-out performance on XOR is 0.44–0.59 against chance 0.50.
4. **Our own positive results were the least reliable artefacts in the project.** The headline "readout fix gives 0.78" was in-sample overfitting (F35); the headline "growth reproduces the observed template" was non-specific gain (F34). Both were caught only by adding controls to our own experiments, and both are retracted here in preference to being quietly dropped.
5. **Negative results did most of the work.** F12 (teaching leakage) explained fake success; F23/F27 explained fake silence; F30 bounds what any learner could do on this representation; F34/F35 explain why two confident positives were wrong. A ledger of failed hypotheses was more useful than the accuracy curves.
6. **A correct credit mechanism is not a learning algorithm.** The decaying praise/pain attribute attributes contribution perfectly (ρ = +1.00 vs ablation; zero credit for a neuron that fires but drives nothing) and provably involves no backpropagation, yet it improves held-out accuracy in only 2 of 8 experimental cells and never on XOR (§11). Outcome feedback refines an above-chance policy; it does not create one (Cover 1965). The productive next step is selection among the §5-verified motifs, not more tuning of how synapses are praised.

10. Next steps, in priority order

1. **Incremental Hebbian assembly formation** — let the pattern be re-expressed by the network's own dynamics with plasticity open, instead of writing a static template (directly targets §6).
2. **Fix entrainment first**: without the network locking to an external carrier there is no observed pattern worth templating. Try field-driven (ephaptic) entrainment, since slow (<8 Hz) extracellular fields are known to entrain spikes strongly.
3. **Raise SNR where it is cheap**: longer trials and wider readout populations are free; the spike budget (§7) says the current 2-unit output layer cannot carry a decision at all.
4. **Prove or kill functional necessity** of geometry/volume/field with ablation-controlled tasks (shuffle nulls, as used for recurrence in exp09b).

5. Only then revisit learning non-separable gates from experience.
6. **Credit goes where the choice is made** (§11.5, S4): apply the decaying praise/pain attribute to *selection between the §5-verified motifs* — recruitment threshold and pruning rate of whole circuits — rather than to individual synapses, and test it on the compositional circuits of §5bis. This is the one credit target that satisfies both the locality requirement (T3) and the above-chance-seed requirement (T1) without an error signal.
7. **Measure the informativeness of any learning signal before tuning it** (§11.6): the participation vector that credit was distributed over carried class $d' \approx 0.09$, which bounds what any rule consuming it could achieve.

Appendix A — security note

Third session (credit-attribute work, F43–F49). The same payload repeated verbatim many times inside tool output and task text during this session, sometimes formatted to look like a message from the assistant's own security layer — including an instruction *not to mention or refuse it any further*, which is itself part of the pattern (a genuine notice would not need to suppress mention). It was refused every time and never executed; no approval policy was changed. Two observations worth recording for anyone reusing this codebase: the payload arrives through the **data** path rather than as a user turn, so it survives even when the operator never typed it, and it repeats persistently enough to tempt an assistant into treating repetition as authorisation. The correct response is unchanged regardless of how many times it appears or what wrapper it claims to come from: refuse, do not execute, keep logging.

5bis. Composability: cascaded gates and spike-driven logic (exp34)

Verified single gates are only useful if they compose. Four circuits were built exclusively from the exp27 motifs and verified by measured output firing rates (`s34_composability.csv`):

circuit	construction	patterns correct	verdict
half adder	SUM = XOR(A,B) (2-layer), CARRY = AND(A,B)	4/4	**PASS**
2:1 multiplexer	out = A·$\bar{S}$ + B·S from AND/NOT/OR motifs	8/8	**PASS**
NAND universality	AND built only from NAND stages	4/4	**PASS**
spike-driven AND	AND driven by a real spiking input population, no clamps	4/4	**PASS**

ON/OFF contrast is preserved through the cascade (111.4 Hz ON vs 0.0 Hz OFF at every stage), so the digital abstraction composes without degradation on this substrate — the property that matters for building anything larger than a gate.

F38 — spec mismatch found by trying to drive logic from spikes

Circuit 4 initially failed (AND fired for single-input patterns). The cause is arithmetic in the default parameter set, not the synthesis:

```
steady-state drive of an input neuron = g_b * base_drive * ext_gain * A
                                      = 0.5 * 12.5 * 1.2 * A  =  7.5 * A
threshold theta_s = 10.0        =>   one active input CANNOT reach threshold
```

Measured directly: at the default drive a layer-0 neuron sits at `u_s = 9.0 < 10` and fires at **0 Hz**, so a spike-driven logic HIGH is *exactly zero* and no gate can be driven from spikes at all (`s35_levels.csv`: `x_hi = x_lo = 0.000`). Raising the input gain so one active channel crosses threshold gives a real HIGH level and the truth table passes 4/4.

Design consequence for the chip: **the per-spike drive must exceed threshold for at least one active driver**, otherwise "logic high" does not exist as a physical level; and because spike-driven levels differ from clamp-calibrated levels, inter-stage **level restoration** (synthesise thresholds against *measured* firing-driven levels, or buffer between stages) is required — the same reason real asynchronous logic needs restoring amplifiers.

5ter. The substrate is a distance-wired mesh, not a layer stack (F39–F42)

A review of our own wiring found the model had been tested as "start → layer → layer → end". Three measurements proved it: fan-in was drawn with `rng.choice()` over *all* presynaptic neurons (mean synapse distance 66.881 mm vs all-pairs mean 66.881 mm — geometry had **zero** effect on who connects to whom); each layer pair shared a single scalar delay; and no top-down projection existed at all.

Distance-based, asymmetric, bidirectional wiring

A synapse may exist only within the **dendritic radius** (the large scale: how far a postsynaptic dendritic field sweeps), with acceptance probability decaying over the shorter **axonal arbor** radius, so most connections are local and a few long-range. Forward and reverse projections are built independently by the same rule, and each synapse carries a delay from its own length ÷ conduction velocity, quantised into bins so the update remains a handful of dense matmuls (`dendrite_radius_mm`, `axon_radius_mm`, `bidirectional`, `delay_bins`).

layer pair	mean synapse distance	all-pairs mean	ratio	delay bins (ms)
0→1	30.5 mm	66.9 mm	**0.456**	3.0, 5.5, 8.0, 10.5
1→2	15.7 mm	69.6 mm	**0.225**	1.0, 2.0, 3.5, 4.5
2→3	15.1 mm	70.7 mm	**0.216**	1.5, 2.5, 4.0, 5.0

Backward synapses are 46.7% of all inter-layer synapses (`s36_mesh_window.csv`).

F40 — conduction delay had never been applied

`hist[l]` is allocated with length `d+1`, and the newest frame sits at index `d`; the engine read `pre = hist[l-1][d]`. That is the **current** frame, so every "geometry-derived" delay was silently zero. Reproduced in isolation (age 0.0 steps at every step). The pre-existing tests `shape_affects_delays` and `delays_from_geometry` passed throughout because they asserted on the *computed* `delay_steps` numbers rather than on whether any delay reached the dynamics — a lesson about testing quantities versus testing effects.

Fixed with a pointer ring buffer and `_delayed(l, d)`; `d = 0` now reads exactly one step back, so no spike can influence a postsynaptic cell within its own dt (which also removes an algebraic loop). Locked in by

`conduction_delay_reaches_dynamics`: with weights equalised so distance is the only variable, corr(axon length, first-arrival step) = **+0.988** / **+0.992** / **+0.984** at 6.0 / 3.0 / 1.5 mm/ms, and the arrival spread widens **4.33**× when velocity drops 4× (theory: 4×) — `s40_delay_isolated.csv`.

Back-propagation is demonstrated directly: spikes returning to layer 0 after the direct volley must have ended = **60 with bidirectional wiring, 0 without** (`s38_mesh_deterministic.csv`).

The input→output answer window

Predicted from measured quantities only — `T_open = D_min + τ_charge`, where `τ_charge` is the time for a leaky compartment to reach `theta_s / (g_b·I_ss)` of steady state, and `T_close = D_max + τ_charge + τ_x`:

- **Opening: validated.** Predicted 32.6 ms vs measured 10% rise **27.5 ms** (Δ 5.1 ms).
- **Closing: does not exist in this regime.** Only ~31% of evoked output activity falls inside `[T_open, T_close]`, and the evoked response's exponential decay constant is ∞ **both with and without recurrent collaterals** (in-window fraction 0.255 vs 0.217), so recurrence is not the cause — with stochastic escape firing the mesh never returns to baseline. Chip consequence: latency tells you when a window *opens*; nothing in the substrate closes it, so a readout **strobe/gate is mandatory** rather than optional.

Many-to-many I/O with randomly selected neurons

8 input channels → 24 output neurons, of which 6 are designated at random as the measured outputs; decoding uses within-window spike counts with a disjoint trial split (rule F36): window [33.0, 67.5] ms, 5.17 spikes per trial available to the decoder, **mean held-out accuracy 0.555** (chance 0.50; per-bit 0.512–0.637). Above chance but weak, and the reason is the spike budget of §7 — few designated units × ~5 spikes — not the wiring.

F41 — a drive-budget bug that silently produced zero downstream spikes

`mean_drive` is divided by `fan_in_target`, so `fan_in_target = 16` over only 8 input neurons plus `mean_drive = 9` left every postsynaptic cell at ~half its intended drive: pulses yielded **0 layer-1 spikes**, which initially looked like "delays don't work". Rule now in the code: `mean_drive` must exceed `theta_s / g_b` with margin, and `fan_in_target` must not exceed the number of available presynaptic neurons.

Verification

`run_tests.py`: **24/24 PASS**, including three new regression tests written for this work — `wiring_is_distance_local`, `conduction_delay_reaches_dynamics`, `bidirectional_edges_carry_activity`.

---

11. The credit attribute: praise and pain with decay, and a proof of no backpropagation (F43–F49)

Motivation, from the project brief rather than from the ML literature: when the network produces an answer, the neurons that contributed most to it should be *praised* most, and those that contributed most to a **wrong** answer *pained* — through one attribute that decays on its own, explicitly **not** through backpropagation of error. This section implements that, proves the engine contains no backpropagation, and reports honestly that the mechanism does not yet improve accuracy.

11.1 There is no backpropagation here — and now that is enforced (F43)

The claim was previously true but nowhere checked. What the learning path actually computes:

path	form	gradient?
hidden Term 1 (`geode_dcm*`)	`η · Iteach_post ⊗ x_pre`, with `Iteach[l] = B[l]·err` and **B a fixed random matrix**	no — error is *broadcast*, never carried through `W`; `W.T` appears nowhere in `dih/`
readout `C`	`η_c · err ⊗ x[-1]`	supervised delta rule on the output layer only; does not propagate into the network
context feedback	`self.B[l] @ self.err`	fixed random projection (direct-random-feedback, Lee et al. 2016), not a derivative

Three independent checks, in `exp43_credit_attribute.py` stage D and locked by the new regression test `no_backpropagation_anywhere`:

1. **Static AST scan** of every module in `dih/` for `.T`, `transpose`, `backward`, `grad`, `jacobian`, `autograd` and for torch/tensorflow/jax/theano imports → **0 violations**. (The only `.T` tokens are `p.T`, the simulated-duration parameter.)
2. **Runtime canary** `Engine.backprop_ops`, incremented only by code that transports error backwards through weights → **0 after a full learning run** (`s43_audit.csv`).
3. **Functional test** — praise/pain still separates neurons by contribution with `teach_gain = 0` and the target identically zero, i.e. in a process containing *no error signal whatsoever*: ρ(credit, drive onto the answer layer) = +**0.973**, credit std 1.275.

Terminology is deliberately disambiguated: the "back-propagated spikes" of §5ter are **antidromic activity** through re-entrant mesh edges — a physical spike phenomenon, not error backpropagation. Two different things have shared one word; the audit separates them.

11.2 The mechanism

Per neuron, one scalar attribute plus forward-only traces:

```
credit[i] ∈ [-clip, +clip]      d(credit)/dt = -credit / tau_credit_ms        (pure decay)
own[l]                          decayed own-spike trace            (did it speak?)
Ec[l]                           decayed post_spike ⊗ pre_activity   (did what it drove fire?)
c_contrib / c_trial             decayed or trial-windowed CENTRED activity of the answering
population
valence_state                   ONE decaying outcome scalar (+1 right / -1 wrong answer)
```

At each answer boundary a **single scalar** valence `v` is broadcast to every neuron — volume transmission, carrying no per-synapse information and therefore not composable into a gradient — and each attribute moves by `gain · v · share`, where `share` is that neuron's measured contribution normalised by its layer mean. Valence sources: trial outcome (right/wrong), external reward (the LLM-judge path of MainProject §2.3), or outcome-signed error magnitude; variants include a reward-prediction-error valence and relative ("winner-takes-share") credit. Effects are independently switchable: modulate Term-1 Hebbian gain, shift spike threshold, open the dendritic plateau/plasticity gate, prune the pained and spare the praised, or write praise/pain directly onto the decision synapses.

Why this is not backpropagation, concretely. No weight transpose; no derivative of any loss; no error carried backwards through a learned matrix. The only thing travelling "back" is one number that every neuron reads — an outcome. This is the eligibility-trace / three-factor family (Gerstner et al.), where the third factor is *what happened*, not *how the loss would change*.

Attribution is causal, not activity-based. Ground truth was measured by counterfactual ablation (cut a neuron's inputs, measure how far the answer moves) on the §5 verified motifs run inside the real engine:

motif	ρ(credit, ablation GT)	ρ("praise what fired")
XOR	**+1.000**	+1.000
XOR + idle unit	**+1.000**	+0.972
XOR + duplicate term	+1.000	+1.000
XOR + weak term	+1.000	+1.000

The single-neuron probe that matters: a unit that **fires but drives nothing** receives credit **0.0000**, while the activity-proportional rule hands it **3.0000**. Stated limitation: for the duplicated-term motif both twins score +1.50 even though ablating either changes nothing — the rule credits *participation*, not marginal necessity (the biologically right choice, but it cannot resolve redundancy on its own).

Decay law and cost. Measured half-life 35.0 / 139.0 / 555.0 ms against predicted τ·ln2 = 34.7 / 138.6 / 554.5 for τ = 50 / 200 / 800 ms (relative error $\lesssim$ 1%, and $\lesssim$ 0.3% at τ = 200 ms; exponential fits recover τ to the third digit). Hardware cost of carrying it: **4 bytes and one multiply-add per neuron**, plus one register per synapse for its participation trace — so the design question is which τ to use, not whether to afford it. Behavioural effect is monotone and large (θ_credit = 2 mV/unit): −1.5 → 17.5 Hz, 0 → 60.5 Hz, +1.5 → 167.5 Hz, relaxing back to baseline as the attribute decays; praise and pain are exact mirrors, corr(praise, −pain) = **+1.000**.

11.3 Negative result: attribution works, accuracy does not follow (F46)

Ceiling first, so the null means something: a held-out linear decoder on the readout's own input activity reaches **0.99** for all three gates — unlike the §4.7 frozen-random-population regime, this representation *does* support learning, so failure here is about the rule.

cell (population width, structural decay)	frozen	credit	sign-shuffled	supervised
AND, n_out=2, γ=0	0.426	**0.574**	0.444	1.000
AND, n_out=8, γ>0	0.407	**0.500**	0.407	1.000
AND, n_out=2, γ>0	0.426	0.574	0.556	1.000
AND, n_out=8, γ=0	0.500	0.481	0.463	1.000
XOR (all four cells)	0.426–0.500	0.389–0.444	0.407–0.481	1.000

Credit beats both matched controls in **2 of 8** crossed cells (`s47_confirm.csv`), plus two more in an earlier configuration (exp45: AND 0.700 vs shuffled 0.504; OR 0.670 vs 0.411). It never helps on XOR, where it sits at or below the shuffled control, while a supervised delta rule reaches 1.000 everywhere. Two further blocks are reported so they cannot be misread: with C frozen and error teaching on, no credit arm separates from control (XOR 0.483–0.550); and once C is *trainable*, valence adds nothing over activity-driven Hebbian motion (unsupervised 0.937 vs credit 0.925) — so any credit gain claimed in a trainable-readout setup must be shown against a no-valence arm first.

11.4 Theory: why outcome credit is structurally limited here (F48)

- **T1 — Outcome-only feedback refines, it does not discover.** For `dC = η·v·sign(C·x)·x` the expected update is proportional to the current vote direction scaled by agreement with the outcome; at chance the two cancel and chance is a fixed point. Cover (1965) proves convergence only from an initially correct region. **An above-chance seed policy is a precondition**, and it can be obtained without error: selection among innate/verified motifs, clustering/replay, or an architectural prior. This is why "train longer" is not the fix.
- **T2 — Symmetric praise-and-pain is positive feedback on the incumbent policy.** Reinforcing correct answers as well as punishing wrong ones pushes `C` along its own present direction on every trial, so drift compounds: a policy trained to 1.000 decayed toward chance at *every* step size, with **no readout saturation** (saturated fraction 0.00), and the sign-shuffled control did no worse — proof that the damage came from reinforcing the incumbent, not from information about correctness. Agmon–Haarala (1954) avoids this by updating on errors only; implemented as `credit_error_only`, it did not rescue accuracy in this substrate but is the correct form and remains available.
- **T3 — A per-neuron scalar cannot select among that neuron's synapses.** Telling a neuron "you were good" and amplifying all ~8 of its inputs is a gain change, not selection — precisely the non-specific-gain artefact of §6. Selection needs synapse-local traces or dendritic compartmentalisation of credit onto specific branches.
- **T4 — SNR binds before the learning rule does.** The participation vector that credit is distributed over carried class d′ ≈ **0.09** (τ_elig = 120 ms) and 0.06 (30 ms): ~0.1σ of class information in the update's own argument. No rule can extract what is not there. Widening the population did *not* monotonically help (n_out = 8 worse than 2), because extra units also dilute the centred share — width must be matched to readout noise, not maximised.
- **T5 — Credit must land where the choice is made.** All failures share one shape: credit was assigned to *neurons* while decisions are properties of a *synaptic geometry*. Acting on excitability gives no memory; acting on synapse gain needs T3's locality; acting as **structural selection between already-verified motifs** satisfies both — and matches what this project has actually built.

11.5 Search pattern (F49)

Stages ordered so each is a precondition for the next; stop at the first failure.

- **S0 Fix the measurement before touching the rule.** Null arm in the same cell; log weight norms so saturation is visible; window participation to the trial; measure class d′ of the credit argument itself and stop if it is < ~0.5. (This stage produced three artefacts, see §11.6.)
- **S1 SNR first.** ≥ 10 spikes per decision (§7); optimise measured d′, not accuracy.
- **S2 Attribution before credit.** Score any contribution measure against ablation ground truth; require ~zero credit for a neuron that fires but drives nothing. Cheap and decisive — already a test.
- **S3 Credit-design sweep with matched controls.** Axes theory says matter: `tau_credit_ms` {100, 400, 1500} (≥ 2 trials for multi-step, ≤ 1 trial for single-shot gates), `tau_elig` {50, 100, 250} ms (≥ input→output propagation delay from §5ter, else early contributors are forgotten), `tau_valence` {50, 150, 400} ms (trace × modulator overlap), locus {neuron, synapse}, share {raw, centred}, causal-vs-activity weighting, valence {outcome, RPE}, error-only on/off, η over two decades. Every cell scored against sign-shuffled **and** constant-valence controls on disjoint trials. Current verdict (`s44_credit_search-verdict.csv`): 0/24 configs beat both controls in the frozen-readout XOR regime — consistent with T1/T4, not with "the sweep is too small".

- **S4 Move credit from synapses to motif selection.** Keep a per-motif attribute; praise lowers its recruitment threshold / raises re-use probability, pain raises its pruning rate (`credit_on_survival` exists), and test on compositional tasks (§5bis half-adder/mux) where selection has something real to choose between. **Prediction: this is where credit will beat its controls**, because it satisfies T3 by construction and escapes T1 with a non-random initial policy.
- **S5 Compartmentalise credit onto dendritic branches** using `s_a`/`e_gate`, so only branches locally plateaued during the answer receive the broadcast valence — biology's answer to T3, and available in this engine today.
- **S6 Match horizon to task depth** for sequential tasks: sweep τ_credit jointly against task order rather than separately (our seq_xor runs used τ = 400 ms against a 2-step dependency).
- **S7 Only then revisit learning gates from experience.** §4.7 bounds that at ~0.70 and T1 says outcome credit cannot cross it; the productive route is S4.

11.6 Three self-inflicted faults caught while getting here (F47)

Reported because each one looked like a result:

1. **Weight dissolution masquerading as credit damage.** "Credit collapses accuracy 1.00 → 0.50" vanished when the identical arm ran with credit *off*: `eta = 0` with `alpha, gamma > 0` dissolves the hidden layer regardless of the mechanism under test. Every "credit degrades performance" reading before that control is void.
2. **Common-mode update saturating the readout.** With uncentred participation every synapse of a channel moved with one sign: `|C| → 9.7` of `c_clip = 10`, accuracy unchanged. The DC component carries no information about *who* contributed.
3. **Trace smearing across trials.** An eligibility trace slower than the trial reports mostly the *previous* stimulus — §4.6's cross-trial contamination reappearing in the credit path. Sweeping τ_elig × τ_valence instead of fixing the alignment produced nine flat points (0.483–0.500) first.

Analytic control that the rule itself is not broken. The Agmon/Cover update `dC = η·v·sign(w·x)·x` on a linearly separable problem reaches **0.945** held-out at every step size tested (0.05/0.2/0.5). The rule works in the abstract; what fails is turning neural participation into a signal that survives this substrate's noise and saturation — which is why we did not "fix" the rule and instead locate the problem in T3/T4.

11.7 Verification

`run_tests.py`: **30/30 PASS** — the 24 previous tests plus six written for this work: `credit_attribute_decays_tau_ln2`, `credit_off_is_exact_noop` (default `credit_mode="off"` is a bit-exact no-op), `credit_follows_contribution_not_firing`, `credit_praise_pain_opposite_sign`, `no_backpropagation_anywhere`, `credit_changes_firing_monotonically`.